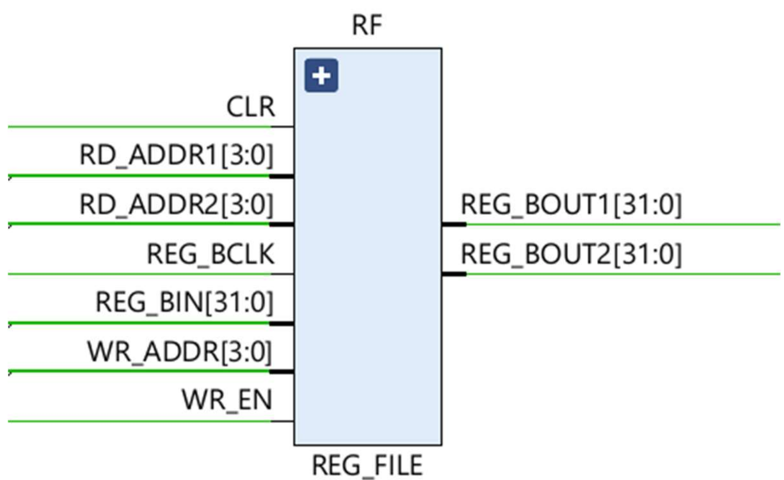
Register File Design Description

Introduction

The register file acts as the primary storage element of the pipelined ALU system. It stores operand values and receives writeback results from completed ALU operations.

Organization



The design contains multiple 32-bit registers addressed through register addresses supplied by the control unit.

Read Ports

Two independent read ports allow simultaneous retrieval of operand A and operand B for ALU execution.

Write Port

A dedicated write port receives the destination address, write enable signal, and result data from the writeback stage.

Read Operation

Read addresses select source registers. The selected register contents are forwarded toward the operand pipeline registers.

Write Operation

On the active clock edge, if write enable is asserted, the incoming result is written into the destination register.

Reset Behavior

Reset logic initializes register contents into a known state, simplifying simulation and verification.

Interaction with Pipeline

The register file provides operands for instruction execution and receives completed results after pipeline latency.

Verification Strategy

Verification checks correct reads, writes, reset behavior, address decoding, and simultaneous read/write functionality.